

			Allow ecf from (a)(i)	
		3	$\frac{1}{2} m v^2 = m \times \Delta \phi_{\text{earth-max}}$ $\frac{1}{2} v^2 = \Delta \phi_{\text{earth-max}} = 61.5 \times 10^6$ $v = 11\,100 \text{ m s}^{-1}$	C1 A1
	(b)	(i)	Gravitational force provides the centripetal force. $\frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2$ $\frac{6.67 \times 10^{-11} (5.98 \times 10^{24})m}{r^2} = mr\left(\frac{2\pi}{27.4 \times 24 \times 3600}\right)^2$ $r = 384\,000 \text{ km}$	M1 M1 A0
		(ii)	Linear speed = $r \omega$ $= 3.84 \times 10^8 \times \left(\frac{2\pi}{27.4 \times 24 \times 3600}\right)$ $= 1020 \text{ m s}^{-1}$	C1 A1
	(c)	1.	Student A's explanation is not valid because both the astronaut and spacecraft will always have weight due to the force of gravity while they are orbiting around the Earth, even if it is small.	B1
		2.	Student B's claim is not correct. The centripetal force is the resultant force which is the force of gravity in this case. <u>Both have different masses hence their centripetal force is different.</u>	B1

3	(a)(i)	Random motion of a large number of molecules in random velocity means no preferred direction of movement or the molecules move in different/all directions with a root-mean-square speed. Momentum is a vector given by the product of mass and velocity. Since the velocities cancel/ or idea of speed in opposite directions, the average velocity is zero and hence the average momentum of the gas molecules of same mass in a container is zero.	M1 A1
	(a)(ii)	When the gas molecules hit the wall of the container, they rebound off the wall with the same speed and experiences a rate of momentum change normal to the wall, implying there is force acting on the molecule by the wall. Other molecules also have a momentum change normal to the wall when they strike and rebound. By Newton's third law, the molecules exert an equal and opposite force on the walls. Since there are a large number of molecules, the number of collisions at any instant in time is very large and practically constant, resulting in a constant gas pressure. The pressure on the walls of the container is the average force per unit area exerted by the molecules on the walls of the container.	B1 B1 B1
	(b)(i)	<u>Increase in internal energy of a system</u> is the <u>sum</u> of the <u>work done on the system</u> and <u>heat supplied to the system</u>.	B1 B1
	(b)(ii)	1. Top surface moves down due to load acting a force on it, hence WD by load on air = $mgx = 500 \times 9.81 \times 0.050 = 245 \text{ J}$ Since system is well insulated, $Q_{to \text{ air}} = 0$ Using the first law of thermodynamics, Increase in $U = Q_{to \text{ air}} + W_{on \text{ air}} = 0 + 245 \text{ J}$ $= 245 \text{ J}$	C1 C1 A1
		2. Using the equation of state for an ideal gas, $p_1 V_1 = nRT_1$ ----- (1) $p_0 V_0 = nRT_0$ ----- (2) (1) – (2): $p_1 V_1 - p_0 V_0 = nRT_1 - nRT_0 = nR (T_1 - T_0)$ Given that $U = \frac{3}{2} nRT$, $p_1 V_1 - p_0 V_0 = \frac{2}{3} \Delta U$ $p_1 (1.947) - (101 \times 10^3)(2.000) = \frac{2}{3} (245)$ $p_1 = 103.8 \text{ kPa} = 104 \text{ kPa}$	C1 C1 A1

1	(a)	(5)(20) + (100)(1.2)	M1
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